UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 8

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To further analyze the least-squares problem, we need the notion of singular values.

- Recall that a matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$ can be decomposed as $\mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$ if $\mathbf{A}$ is diagonalizable.
- What if $\mathbf{A} \in \mathbb{R}^{m\times n}$ with $m \neq n$ or $\mathbf{A}$ is not diagonalizable?

Theorem 1 (SVD[1] theorem)**.** *Given an arbitrary matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, there exist matrices $\mathbf{U} \in \mathbb{R}^{m\times m}$, $\mathbf{V} \in \mathbb{R}^{n\times n}$ and $\mathbf{\Sigma} \in \mathbb{R}^{m\times n}$, such that:*

(i) $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$.

(ii) $\mathbf{U}$ *and* $\mathbf{V}$ *are orthogonal matrices (*$\mathbf{U}^{\mathrm{T}}\mathbf{U} = \mathbf{I}_m$, $\mathbf{V}^{\mathrm{T}}\mathbf{V} = \mathbf{I}_n$*).*

(iii) $\mathbf{\Sigma}$ *is in the form of*

$$\left[\begin{array}{ccc|ccc} \sigma_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_m & 0 & \cdots & 0 \end{array}\right]$$

$$\underbrace{\qquad\qquad}_{m\times m} \quad \underbrace{\qquad\qquad}_{m\times(n-m)}$$

if $n \geq m$, and in the form of

$$\left[\begin{array}{ccc} \sigma_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \sigma_n \\ 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 0 \end{array}\right] \begin{array}{l} \left.\begin{array}{c}\\\\\\\end{array}\right\} n\times n \\ \left.\begin{array}{c}\\\\\\\end{array}\right\} (m-n)\times n \end{array}$$

if $n \leq m$, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

Here, $\sigma_1, \sigma_2, \ldots$ are called the *singular values* of $\mathbf{A}$.

- Let r be the number of non-zero singular values of $\mathbf{A}$:

$$\underbrace{\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r}_{\text{non-zero}} > \sigma_{r+1} = \sigma_{r+2} = \cdots = 0.$$

  Theorem: $r = \mathrm{rank}(\mathbf{A})$.

- $\mathbf{\Sigma}$ is called a rectangular diagonal matrix.
- If $\mathbf{A}$ is symmetric, we have the eigenvalue decomposition $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{\mathrm{T}}$. Since the diagonal of $\mathbf{\Lambda}$ may not be positive, $\mathbf{\Sigma}$ is not the same as $\mathbf{\Lambda}$.
- If $\mathbf{A}$ is symmetric and PSD, then $\mathbf{\Lambda} \geq 0$. We can select $\mathbf{\Sigma} = \mathbf{\Lambda}$ and $\mathbf{V} = \mathbf{U}$, then the eigenvalues and singular values are the same.

[1] SVD refers to singular value decomposition.

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Moore-Penrose inverse. Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$. Define the Moore-Penrose inverse or pseudoinverse of $\mathbf{A}$ as

$$\mathbf{A}^{+} = \mathbf{V}\mathbf{\Sigma}^{+}\mathbf{U}^{\mathrm{T}}$$

(note: $\mathbf{U}^{-1} = \mathbf{U}^{\mathrm{T}}$, $\mathbf{V}^{-1} = \mathbf{V}^{\mathrm{T}}$), where

$$\mathbf{\Sigma}^{+} = \begin{bmatrix} \frac{1}{\sigma_1} & & & & & 0 & \cdots & 0 \\ & \ddots & & & & & & \\ & & \frac{1}{\sigma_r} & & & & & \\ & & & 0 & & & & \\ & & & & \ddots & & & \\ & & & & & 0 & 0 & \cdots & 0 \end{bmatrix}$$

if $n \leq m$, and

$$\mathbf{\Sigma}^{+} = \begin{bmatrix} \frac{1}{\sigma_1} & & & & & & \\ & \ddots & & & & & \\ & & \frac{1}{\sigma_r} & & & & \\ & & & 0 & & & \\ & & & & \ddots & & \\ & & & & & 0 & \\ 0 & & & & & 0 & \\ \vdots & & & & & \vdots & \\ 0 & & & & & 0 & \end{bmatrix}$$

if $m \leq n$.
(So, we take inverse of only the *positive* singular values.)

Example 1. Let $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, then we have $\mathbf{U} = \mathbf{V} = \mathbf{I}_2$ and $\mathbf{\Sigma} = \mathbf{A}$. The Moore-Penrose inverse of $\mathbf{A}$ is given by

$$\mathbf{A}^{+} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix}.$$

(Here $\sigma_1 = 2$, $\sigma_2 = 0$.) ■

If $\mathbf{A}$ is invertible (thus must be square) and $\sigma_1, \ldots, \sigma_n \neq 0$, then $\mathbf{\Sigma}^{+} = \mathbf{\Sigma}^{-1}$. In this case, we have

$$\mathbf{A}^{-1} = \left(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}\right)^{-1} = \left(\mathbf{V}^{\mathrm{T}}\right)^{-1}\mathbf{\Sigma}^{-1}\mathbf{U}^{-1} = \mathbf{V}\mathbf{\Sigma}^{+}\mathbf{U}^{\mathrm{T}} = \mathbf{A}^{+}.$$

If $\mathbf{A}$ is invertible, then $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_n$. However, in general, $\mathbf{A}\mathbf{A}^{+} \neq \mathbf{I}_n$.

Example 2. In the previous example,

$$\mathbf{A}\mathbf{A}^{+} = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} \frac{1}{2} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \neq \mathbf{I}_2.$$

■

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- If $\mathbf{A} \in \mathbb{R}^{m\times n}$ is a fat matrix with linearly independent rows ($m \le n$):

$$\mathbf{A} = \mathbf{U}\begin{bmatrix}\sigma_1 & & & 0 & \cdots & 0\\ & \ddots & & \vdots & & \vdots\\ & & \sigma_m & 0 & \cdots & 0\end{bmatrix}\mathbf{V}^{\mathrm{T}},$$

  in this case, $\mathbf{A}^{+} = \mathbf{A}^{\mathrm{T}}(\mathbf{A}\mathbf{A}^{\mathrm{T}})^{-1}$.

- If $\mathbf{A} \in \mathbf{R}^{m\times n}$ is a tall matrix with linearly independent columns ($n \le m$):

$$\mathbf{A} = \mathbf{U}\begin{bmatrix}\sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0\end{bmatrix}\mathbf{V}^{\mathrm{T}},$$

  in this case, $\mathbf{A}^{+} = (\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}\mathbf{A}^{\mathrm{T}}$.

Recall that for the least-squares problem

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|,$$

the set of solutions is given by

$$\mathcal{S} = \left\{\boldsymbol{x}^* \mid \mathbf{A}^{\mathrm{T}}\mathbf{A}\boldsymbol{x}^* = \mathbf{A}^{\mathrm{T}}\boldsymbol{y}\right\}.$$

Corollary 6.1 in the textbook (see the proof in textbook):

Corollary 1. *(i)* $\mathcal{S} = \mathbf{A}^{+}\boldsymbol{y} + \mathcal{N}(\mathbf{A})$.

(ii) The solution to $\min\|\boldsymbol{x}\|$ *subject to* $\boldsymbol{x} \in \mathcal{S}$ *(minimum length solution to LS) is unique and equal to* $\mathbf{A}^{+}\boldsymbol{y}$.

Interpretation: Solution to LS may not satisfy $\mathbf{A}\boldsymbol{x} = \boldsymbol{y}$, but $\mathbf{A}^{+}\boldsymbol{y}$ is the minimum norm solution to LS, and every other solution is this minimum norm solution plus a vector in the null space of $\mathbf{A}$.

- If $\mathbf{A}$ has full column rank, then $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$, and the minimum norm solution is also the only solution to LS:

$$\mathbf{A}^{+}\boldsymbol{y} = \left(\mathbf{A}^{\mathrm{T}}\mathbf{A}\right)^{-1}\mathbf{A}^{\mathrm{T}}\boldsymbol{y}.$$

SVD plays a role in defining $\mathbf{A}^{+}$.

- Let's revisit SVD:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}} \quad \Rightarrow \quad \mathbf{A}\mathbf{A}^{\mathrm{T}} = \mathbf{U}\mathbf{\Sigma}\underbrace{\boxed{\mathbf{V}^{\mathrm{T}}\mathbf{V}}}_{\mathbf{I}}\mathbf{\Sigma}^{\mathrm{T}}\mathbf{U}^{\mathrm{T}} = \mathbf{U}(\mathbf{\Sigma}\mathbf{\Sigma}^{\mathrm{T}})\mathbf{U}^{\mathrm{T}}.$$

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- If $m \leq n$,

$$\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathrm{T}} = \begin{bmatrix} \sigma_1 & & & 0 & \cdots & 0 \\ & \ddots & & \vdots & & \vdots \\ & & \sigma_m & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_m \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_m^2 \end{bmatrix}.$$

- If $m > n$,

$$\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathrm{T}} = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \sigma_1 & & & 0 & \cdots & 0 \\ & \ddots & & \vdots & & \vdots \\ & & \sigma_n & 0 & \cdots & 0 \end{bmatrix} = \begin{bmatrix} \sigma_1^2 & & & & & \\ & \ddots & & & & \\ & & \sigma_n^2 & & & \\ & & & 0 & & \\ & & & & \ddots & \\ & & & & & 0 \end{bmatrix}.$$

  (A diagonal matrix with $\sigma_1^2, \ldots, \sigma_n^2$, and $m - n$ zeros on the diagonal.)

- Therefore, $\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{\mathrm{T}}$ is always diagonal, and

$$\mathbf{A}\mathbf{A}^{\mathrm{T}} = \mathbf{U} \begin{bmatrix} \text{diagonal matrix} \\ \text{with non-negative} \\ \text{diagonal entries} \end{bmatrix} \mathbf{U}^{\mathrm{T}}.$$

  Also, $\mathbf{A}\mathbf{A}^{\mathrm{T}} \succeq \mathbf{0}$.

- This is an eigenvalue decomposition of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$. The columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$, and the eigenvalues of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ are given by

$$\begin{cases} \sigma_1^2, \ldots, \sigma_m^2, & \text{if } m \leq n, \\ \sigma_1^2, \ldots, \sigma_n^2, \underbrace{0, \ldots, 0}_{m-n}, & \text{if } m > n. \end{cases}$$

- Similarly,

$$\mathbf{A}^{\mathrm{T}}\mathbf{A} = \mathbf{V} \left[\text{diagonal matrix}\right] \mathbf{V}^{\mathrm{T}}$$

  is an eigenvalue decomposition of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. The columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$, and the eigenvalues of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ are given by

$$\begin{cases} \sigma_1^2, \ldots, \sigma_n^2, & \text{if } n \leq m, \\ \sigma_1^2, \ldots, \sigma_m^2, \underbrace{0, \ldots, 0}_{n-m}, & \text{if } n > m. \end{cases}$$

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Recap:

- $\mathbf{A}$ can be written as $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$. The columns of $\mathbf{U}$ (called the *left singular vectors*) and the columns of $\mathbf{V}$ (called the *right singular vectors*) are the eigenvectors of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{A}$.
- The non-zero singular values of $\mathbf{A}$ are the square-roots of the non-zero eigenvalues of $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ or $\mathbf{A}^{\mathrm{T}}\mathbf{A}$.

SVD is important in low-rank matrix approximation.

Example 3. Consider k (e.g., one million) pictures, each with n^2 (e.g., $n = 5000$) pixels where the color of each pixel is represented by a number from 0 (black) to 255 (white). This is called grayscale.

- To store these pictures or send them over communication channels, we need to handle a dataset with $k \times n^2$ elements (e.g., $1,000,000 \times 5,000 \times 5,000$).
- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ denote the matrix for picture 1.
- Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$ be the SVD of $\mathbf{A}$.
- Let $\boldsymbol{u}^{(1)}, \ldots, \boldsymbol{u}^{(n)}$ denote the columns of $\mathbf{U}$ and $\boldsymbol{v}^{(1)}, \ldots, \boldsymbol{v}^{(n)}$ denote the columns of $\mathbf{V}$.
- Let r denote the rank of $\mathbf{A}$.
- By expanding $\mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$, we conclude that

$$\mathbf{A} = \sigma_1 \left(\boldsymbol{u}^{(1)}\right)\left(\boldsymbol{v}^{(1)}\right)^{\mathrm{T}} + \sigma_2 \left(\boldsymbol{u}^{(2)}\right)\left(\boldsymbol{v}^{(2)}\right)^{\mathrm{T}} + \cdots + \sigma_r \left(\boldsymbol{u}^{(r)}\right)\left(\boldsymbol{v}^{(r)}\right)^{\mathrm{T}}.$$

- Let's approximate $\mathbf{A}$ by $\sigma_1(\boldsymbol{u}^{(1)})(\boldsymbol{v}^{(1)})^{\mathrm{T}}$.
- Storing/transmitting $\sigma_1(\boldsymbol{u}^{(1)})(\boldsymbol{v}^{(1)})^{\mathrm{T}}$ can be done by putting all the data into a single vector $\begin{bmatrix} \sigma_1 \\ \boldsymbol{u}^{(1)} \\ \boldsymbol{v}^{(1)} \end{bmatrix} \in \mathbb{R}^{2n+1}$, while $\mathbf{A}$ requires $n \times n$ elements.
- So, we compress data from $k \times n^2$ to $k(2n+1)$ which is much lower.

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Note: since $\sigma_1 \geq 0$,

$$\sigma_1 \left(\boldsymbol{u}^{(1)}\right)\left(\boldsymbol{v}^{(1)}\right)^{\mathrm{T}} = \left(\sqrt{\sigma_1}\boldsymbol{u}^{(1)}\right)\left(\sqrt{\sigma_1}\boldsymbol{v}^{(1)}\right)^{\mathrm{T}}. \tag{1}$$

By denoting $\boldsymbol{x}$ as $\sqrt{\sigma_1}\boldsymbol{v}^{(1)}$ and $\boldsymbol{y}$ as $\sqrt{\sigma_1}\boldsymbol{v}^{(1)}$, we approximated $\mathbf{A}$ by $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$.

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Question: If $\mathbf{A}$ is to be approximated as an outer product of two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ like $\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}$, is Equation (1) the best solution or are there better values for $\boldsymbol{x}$ and $\boldsymbol{y}$?

- Define an optimization:

$$\min_{\boldsymbol{x},\boldsymbol{y}} \quad \text{some function of the residue } \left(\mathbf{A} - \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}}\right).$$

- For LS, we had the residue $\mathbf{A}\boldsymbol{x} - \boldsymbol{y} \in \mathbb{R}^m$ and minimized the L_2-norm: $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{y}\|_2$.
- For the residue $\mathbf{A} - \boldsymbol{x}\boldsymbol{y}^{\mathrm{T}} \in \mathbb{R}^{m\times n}$, we need a matrix norm.
- Recall the definition of a norm: $\|\cdot\|: \mathbb{R}^{m\times n} \to \mathbb{R}$, with three properties:
  (1) $\|\mathbf{X}\| \geq 0,\ \forall \mathbf{X} \in \mathbb{R}^{m\times n}$ and $\|\mathbf{X}\| = 0$ iff $\mathbf{X} = \mathbf{0}$.
  (2) $\|\alpha\mathbf{X}\| = |\alpha| \cdot \|\mathbf{X}\|, \forall \alpha \in \mathbb{R},\ \forall \mathbf{X} \in \mathbb{R}^{m\times n}$.
  (3) $\|\mathbf{X} + \mathbf{Y}\| \leq \|\mathbf{X}\| + \|\mathbf{Y}\|, \forall \mathbf{X}, \mathbf{Y} \in \mathbb{R}^{m\times n}$.
- A natural way to define a matrix norm is to reshape it into a vector by stacking all columns.

  Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix} \quad \longrightarrow \quad \mathrm{vec}(\mathbf{A}) = \begin{bmatrix} 1 \\ 2 \\ \vdots \\ 6 \end{bmatrix},$$

  where vec stands for vectorization.
- Define $\|\mathbf{A}\|$ to be $\|\mathrm{vec}(\mathbf{A})\|_2$ (the length of $\mathrm{vec}(\mathbf{A})$).
- This norm is called the Frobenius norm: $\|\mathbf{A}\|_{\mathrm{F}}$:

$$\|\mathbf{A}\|_{\mathrm{F}} = \|\mathrm{vec}(\mathbf{A})\|_2 = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2} = \sqrt{\mathrm{tr}(\mathbf{A}\mathbf{A}^{\mathrm{T}})}.$$

  Here tr stands for trace, the sum of diagonal elements of a matrix.
- Recall the L_p-norm for vectors: for $1 \leq p < \infty$,

$$\|\boldsymbol{x}\|_p = \left(|x_1|^p + \cdots + |x_n|^p\right)^{\frac{1}{p}}, \quad \forall \boldsymbol{x} \in \mathbb{R}^n.$$

- L_p-norm for vectors induces a norm for matrices:

$$\|\mathbf{A}\|_p = \max_{\substack{\boldsymbol{z}\neq\mathbf{0} \\ \boldsymbol{z}\in\mathbb{R}^n}} \frac{\|\mathbf{A}\boldsymbol{z}\|_p}{\|\boldsymbol{z}\|_p}.$$

- It can be shown that $\|\mathbf{A}\|_p$ is a norm (satisfies three properties).

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- $\|\mathbf{A}\|_p$ is called the L_p-induced norm.
- Note: $\mathbf{A}\boldsymbol{z}$ and $\boldsymbol{z}$ are vectors and so we take their L_p-norm for vectors, divide them and maximize over $\boldsymbol{z}$.
- Note that

$$\begin{aligned}\frac{\|\mathbf{A}\boldsymbol{z}\|_p}{\|\boldsymbol{z}\|_p} &= \left\|\frac{1}{\|\boldsymbol{z}\|_p}\mathbf{A}\boldsymbol{z}\right\|_p \\ &\qquad \text{(using property (2) for } \alpha = 1/\|\boldsymbol{z}\|_p\text{)} \\ &= \left\|\mathbf{A}\frac{\boldsymbol{z}}{\|\boldsymbol{z}\|_p}\right\|_p = \|\mathbf{A}\boldsymbol{w}\|_p\,,\end{aligned}$$

  where $\boldsymbol{w} \in \mathbb{R}^n$ and $\|\boldsymbol{w}\|_p = 1$. Therefore,

$$\|\mathbf{A}\|_p = \max_{\|\boldsymbol{w}\|_p=1} \|\mathbf{A}\boldsymbol{w}\|_p.$$

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